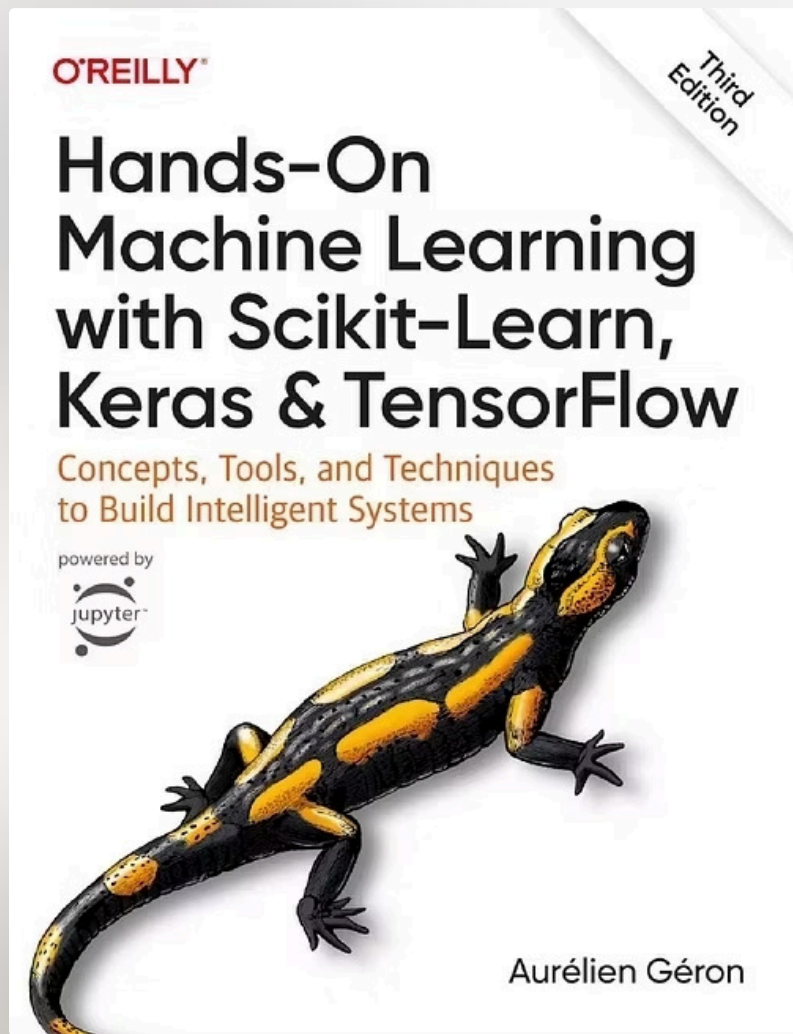




# Machine Learning

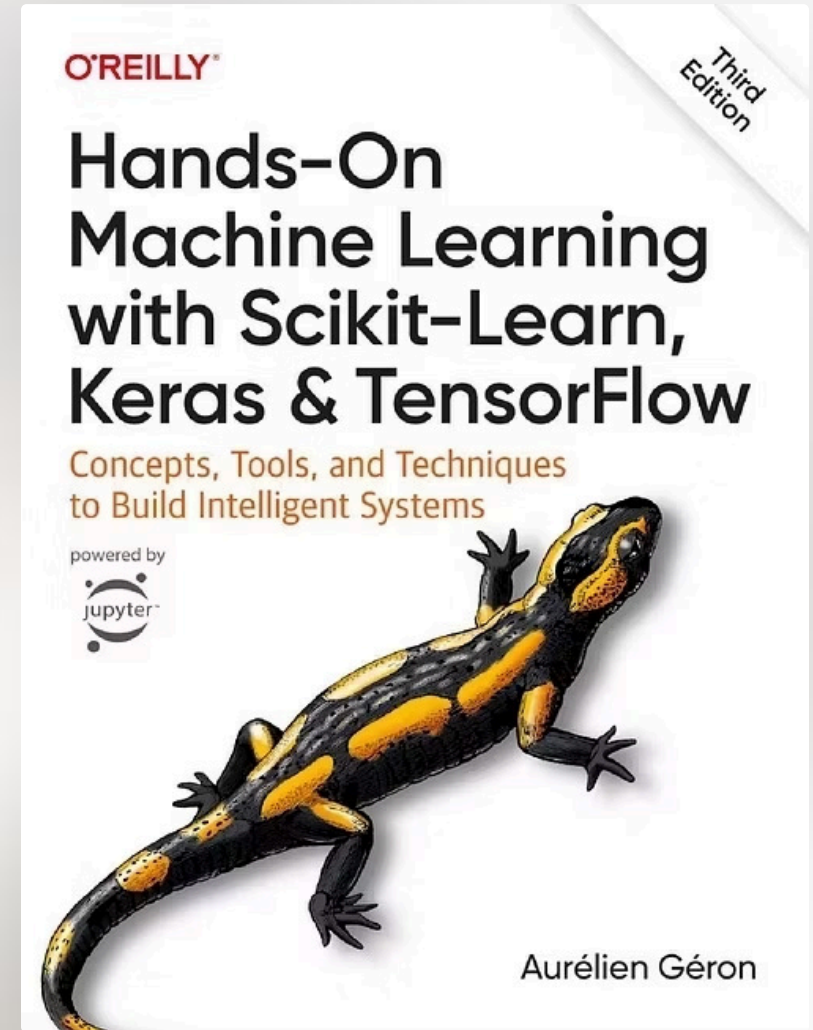
Lessons based on the Textbook "Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow" by Aurélien Géron, 2023.

**RP** by Rodrigo Martins Pagliares

Last update: 07/04/2025

# Machine Learning

## Chapter 1 – The Machine Learning Landscape





# Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
5. Main Challenges of Machine Learning
6. Testing and Validating

# Agenda for Chapter 1

1. What Is Machine Learning?
2. Why Use Machine Learning?
3. Examples of Applications
4. **Types of Machine Learning Systems**
  - **Classification criteria #1: Supervision during training**
    - Supervised, **Unsupervised**, Semi-Supervised, Self-Supervised, and Reinforcement **Learning**
  - **Classification criteria #2: Approach to Learning Method**
    - Batch and Online (Incremental) Learning
  - **Classification criteria #3: Learning Method**
    - Instance-Based and Model-Based Learning
5. **Main Challenges of Machine Learning**
  - Insufficient Quantity of Training Data
  - Non-representative Training Data
  - Poor-Quality Data
  - Irrelevant Features
  - Overfitting the Training Data
  - Underfitting the Training Data
  - Stepping Back
6. **Testing and Validating**
  - Hyperparameter Tuning and Model Selection
  - Data Mismatch



# Classification criteria #1: Supervision during training

Supervised, Unsupervised, Semi-Supervised, Self-Supervised, and Reinforcement Learning

# Unsupervised Learning: Unlabeled Data Set

- Suppose you have a lot of data about your blog's visitors.
  - The training data is unlabeled
- The system tries to learn without a teacher (supervisor)

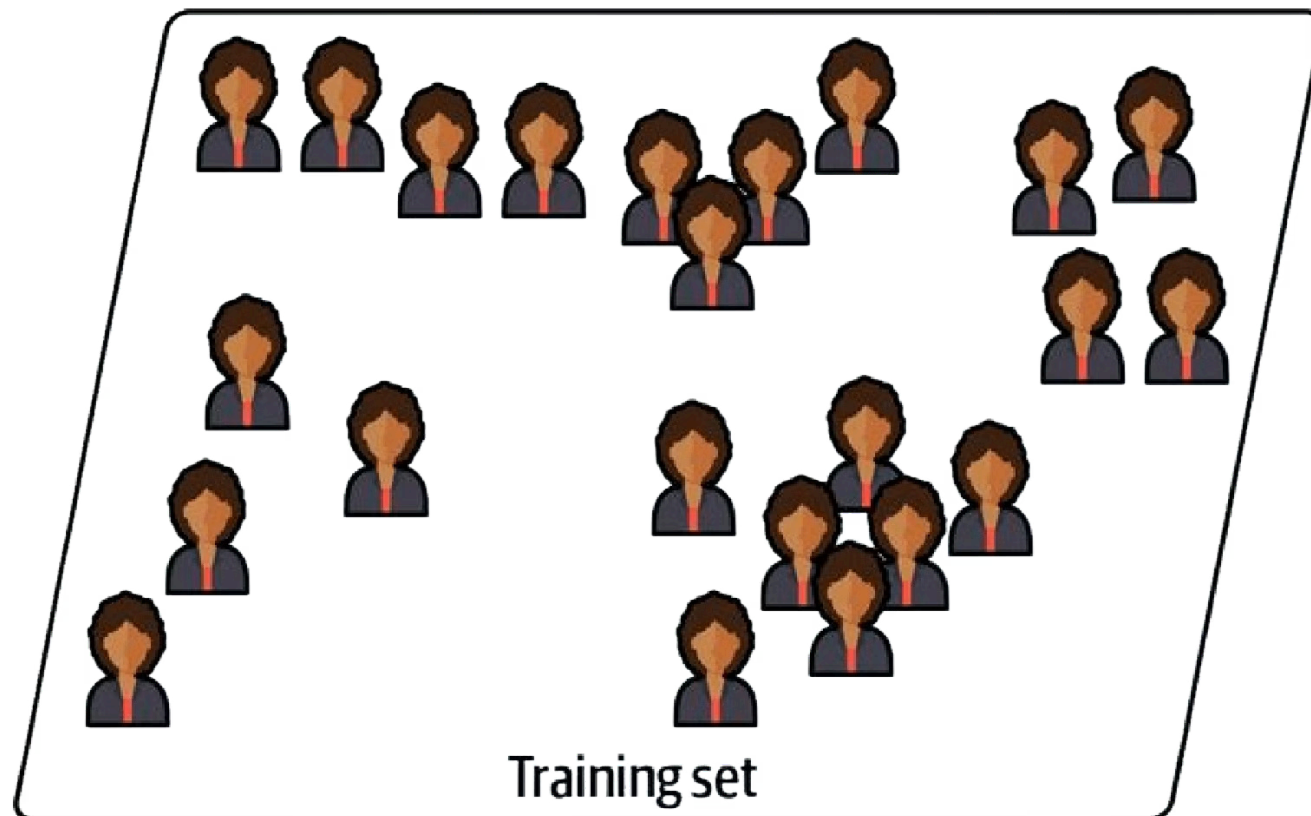


Figure 1-7. An unlabeled training set for unsupervised learning

# Unsupervised Learning: Clustering Task

- You may want to run a **clustering algorithm** on the **unlabeled training data** to try to detect **groups of similar visitors**
- At **no point** do you **tell the algorithm** which **group** a **visitor** belongs to:
  - The **algorithm finds those connections without your help**.
- **Example:** Clustering algorithms **group blog visitors** into **segments** like **teenagers** or **adults** based on **behavior**.

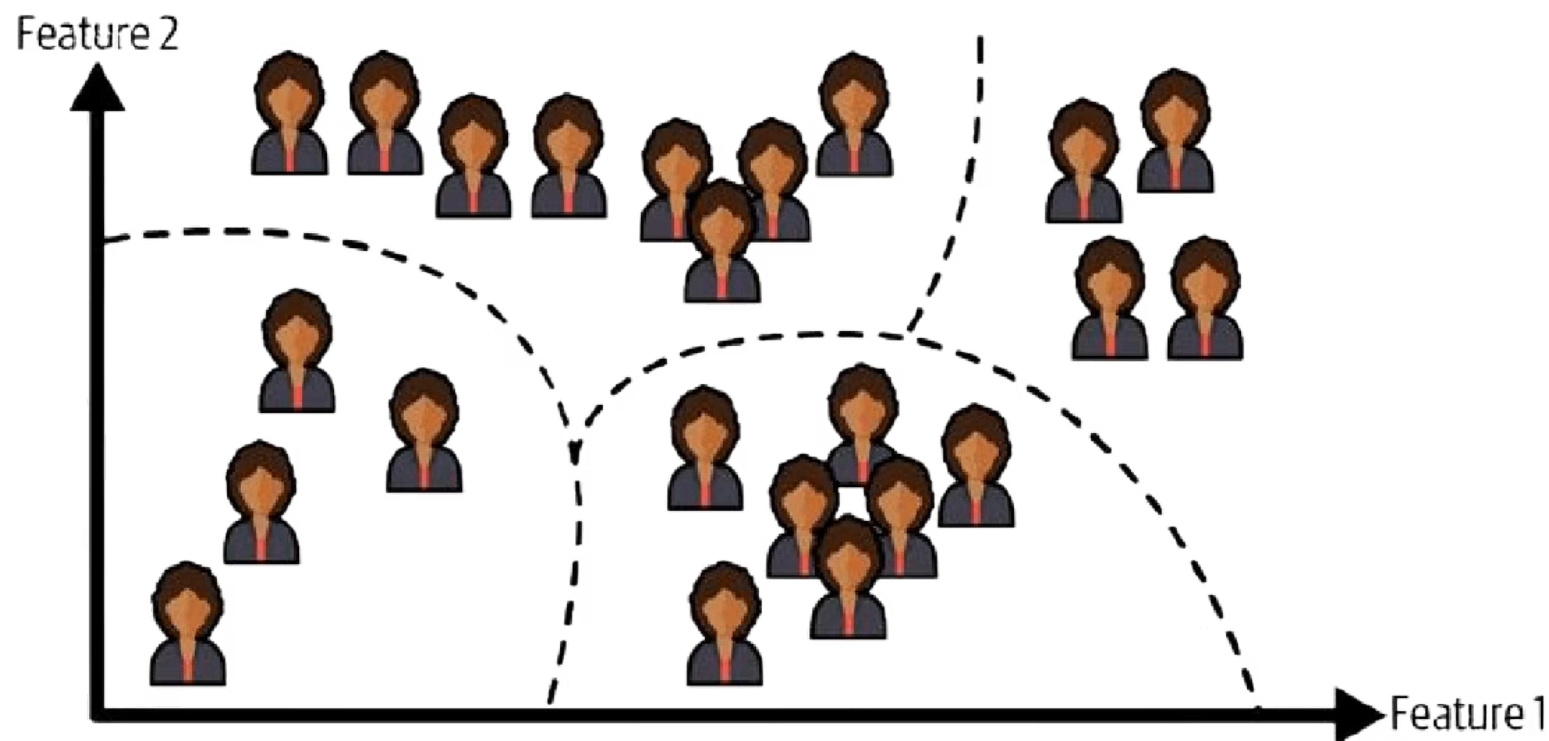


Figure 1-8. Clustering

# Unsupervised Learning: Visualization Algorithms

- You feed the **Visualization algorithms** with a lot of **complex** and **unlabeled data**
  - The **algorithms** output a **2D** or **3D** representation of your data.

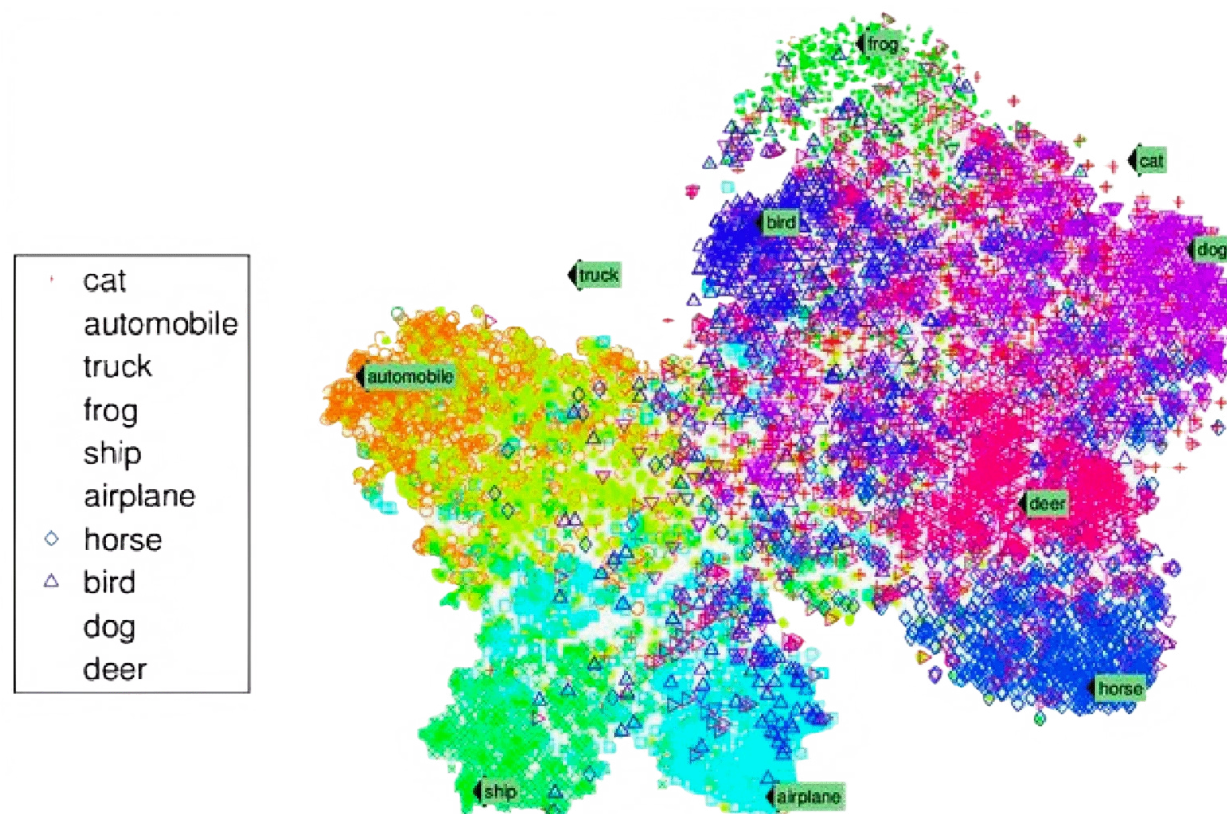


Figure 1-9. Example of a t-SNE\* visualization highlighting semantic clusters

\* t-distributed stochastic neighbor embedding (t-SNE) is a statistical method for visualizing high-dimensional data by giving each datapoint a location in a two or three-dimensional map.

- **High-dimensional** = data with many features (dimensions).



# Unsupervised Learning: Dimensionality Reduction Algorithms

- **Definition:**
  - **Simplifies the data by reducing the number of features or dimensions**
  - **Aims to preserve as much relevant information as possible**
- **Benefits:**
  - **Speeds up training**
  - **Reduces memory and disk space usage**
  - **May improve performance in some cases**
  - **Make the learning process faster and more efficient**

# Feature Extraction in Dimensionality Reduction Algorithms

- **Definition:**
  - **Merges correlated features into one**
  - **Example:** Car mileage and age can be merged into a single feature representing wear and tear
- **Goal:**
  - **Reduce the number of features without losing significant information**





# Why Use Dimensionality Reduction Algorithms?

- It is often a good idea to try to reduce the number of dimensions in your training data using a dimensionality reduction algorithm before you feed it to another ML algorithm
  - Such as a supervised learning algorithm

# Unsupervised Learning: Anomaly Detection Algorithms

- **Definition:**
  - Detects unusual or rare instances in the data
- **How it works:**
  - Trained on mostly normal instances to recognize typical patterns
  - Identifies whether new data points are normal or anomalous



# Unsupervised Learning: Anomaly Detection Algorithms

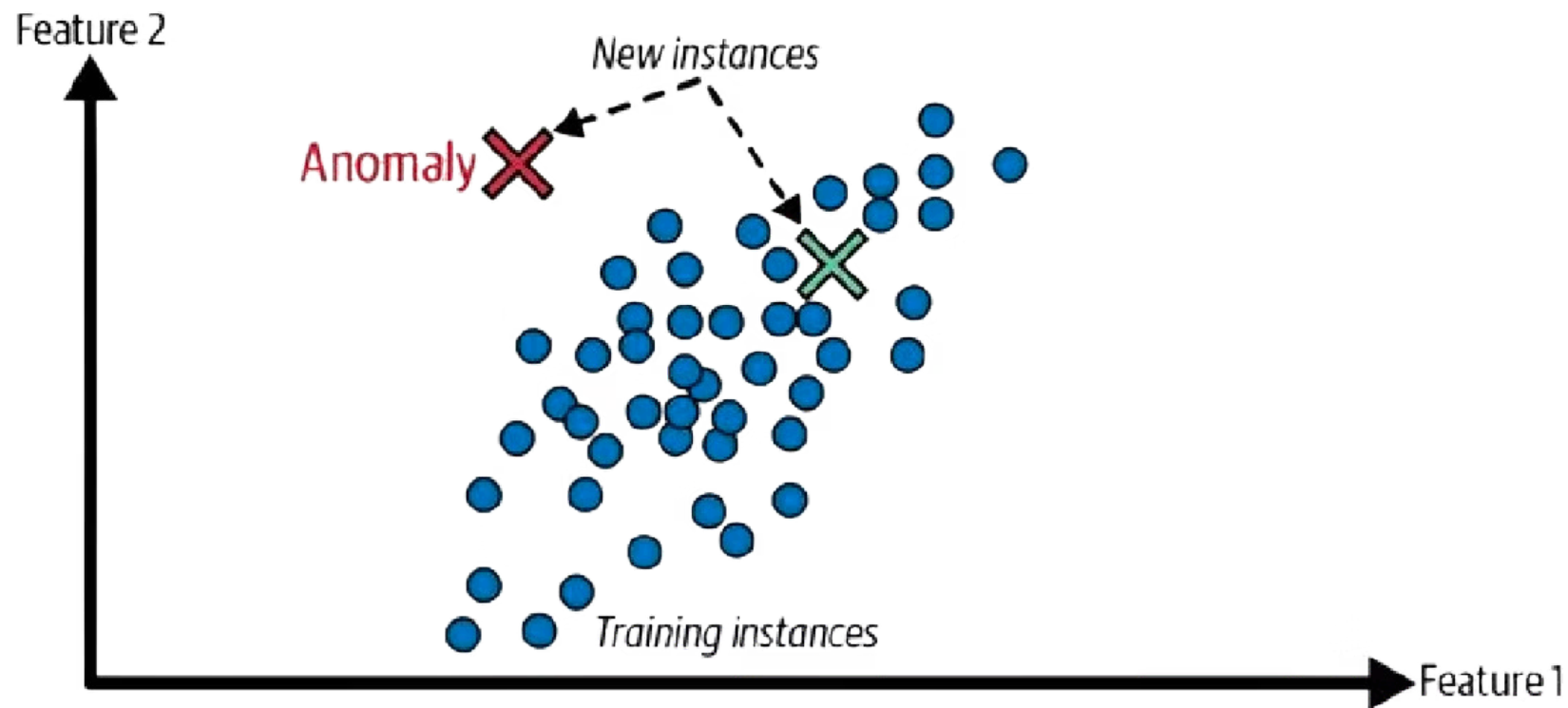


Figure 1-10. Anomaly detection

# Applications of Anomaly Detection Algorithms

- **Credit Card Fraud Detection:**
  - Identify unusual spending patterns
- **Manufacturing Defects:**
  - Spot production errors or defective items
- **Outlier Removal:**
  - Clean the dataset before feeding it into other algorithms



# Unsupervised Learning: Novelty Detection Algorithms

- **Definition:**
  - Detects new instances that differ from all previous instances in the training set
  - Requires a "clean" training set with NO instances that should be detected as novelties
- **Example:**
  - Recognizing new dog breeds in a dataset of thousands of dog pictures



# Difference Between Anomaly and Novelty Detection Algorithms



## Anomaly Detection:

- Identifies **RARE** instances that deviate from normal patterns
  - e.g., unusual Chihuahuas



## Novelty Detection:

- Detects **NEW** instances that differ from the known data **without including them in training**
  - e.g., new breed of dog not seen in the training data

# Unsupervised Learning: Association Rule Learning Algorithms

- **Goal:**
  - Dig into large amounts of data and discover interesting relations between attributes.
- **Example:**
  - Suppose you own a **supermarket**.
  - An **association rule** on your **sales logs** may reveal that **people** who purchase **barbecue sauce** and **potato chips** also tend to buy **steak**.
    - Thus, you may want to **place these items close** to one **another**.



# Summary: Unsupervised Learning Tasks/Algorithms

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## Clustering

Grouping similar data points (e.g., customer segmentation)

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## Dimensionality Reduction

Reducing the number of features while preserving essential information (e.g., **PCA** - Principal Component Analysis)

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## Anomaly Detection

Identifying rare or unusual data points (e.g., fraud detection)

---

## Novelty Detection

Aims to detect new instances that look different from all instances in the training set.

In unsupervised learning, the system learns from **unlabeled data** to find **patterns**.

# Next: Semi-Supervised Learning



# Q&A



## Open Floor

Open floor for discussion or clarifications



## Discussion

Share your thoughts and questions about AI concepts